

From Movement Organization to Whole-Organism Adaptation: An Integrated Framework for Multisystem Physiological Research

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This manuscript reports no original experimental data.

Abstract

Regular physical activity is an established component of the prevention and management of noncommunicable diseases, while contemporary systems and Network Physiology approaches demonstrate that movement can be associated with coordinated responses across multiple physiological systems. The present conceptual paper asks a narrower question: how does the organization of movement relate to the dynamic multisystem response and subsequent adaptation of the whole organism? We propose an integrated framework in which movement is considered both an observable expression of ongoing biological organization and a recurrent perturbation of the internal biological environment. The framework distinguishes movement organization from movement modality, external stimulus, functional performance, acute response, recovery and longer-term adaptation. It further proposes a temporally structured sequence linking movement organization, multisystem response, recovery, repeated exposure, adaptive trajectory and health relevance. Six research propositions define progressively downstream hypotheses concerning acute multisystem response, within-person adaptation, incremental information from temporal movement characteristics, response-guided modulation, health-related outcomes and longitudinal relevance to noncommunicable disease prevention. The paper is conceptual and reports no original experimental data. Its purpose is not to propose a new biological law, diagnostic method or therapeutic modality, but to provide a falsifiable research framework through which the relationship between movement organization, multisystem physiology and whole-organism adaptation can be empirically investigated.

Keywords: movement organization; multisystem physiological response; whole-organism adaptation; integrative physiology; Network Physiology; movement quality; dynamic biological state; noncommunicable disease prevention

1. Introduction

Noncommunicable diseases (NCDs) remain a major global health burden. Cardiovascular diseases, cancers, chronic respiratory diseases and diabetes account for a substantial proportion of premature NCD mortality, while insufficient physical activity is recognized as an important modifiable risk factor. The World Health Organization identifies regular physical activity as an important component of the prevention and management of cardiovascular disease, diabetes, several cancers and other adverse health outcomes (World Health Organization, 2024; World Health Organization, 2025).

The established evidence provides a strong foundation for considering movement as an important determinant of health. Physical activity encompasses a broad range of bodily movements, and its health effects are commonly characterized by attributes such as frequency, duration and intensity. This framework has substantial value for population-level prevention and for quantifying exposure to physical activity. It does not, however, necessarily describe how a movement stimulus is organized by an individual organism or whether different organizations of a comparable movement stimulus may produce different biological responses.

Movement engages interacting neural, sensory, musculoskeletal and physiological processes across multiple biological systems. Contemporary systems and Network Physiology approaches provide established methods for investigating such interactions across different physiological

domains and temporal scales (Kitano, 2002; Bartsch et al., 2015; Balagué et al., 2020; Balagué et al., 2022).

These developments establish that organismal physiology cannot be adequately understood by considering biological systems only as isolated components. The present study does not challenge this established view. Instead, it asks a narrower question: how does the organization of the movement stimulus relate to the subsequent multisystem response and adaptive trajectory of the organism?

A related development has emerged in contemporary integrative physiology. Kuang (2026) proposes a five-domain functional framework intended to provide a coherent and operational representation of whole-body function across multiple levels of organization. The existence of such frameworks reinforces an important point for the present study: the problem is not an absence of scientific knowledge concerning whole-organism integration, but the need to examine how movement itself participates in the organization of the resulting systemic response (Kuang, 2026).

Movement occupies a distinctive position in this problem because it is simultaneously an observable expression of ongoing biological organization and a recurrent perturbation of the organism. Its execution depends on sensory information, neural processing, motor activation, mechanical interaction and changing physiological conditions. The resulting movement is therefore an observable output of the organism. At the same time, movement modifies the internal environment by changing mechanical loading, energy demand, sensory input, cardiorespiratory requirements, metabolic activity and other physiological processes.

This dual role creates a distinction between the form of movement and the organization of the biological stimulus experienced by the organism. Walking, swimming, lifting, jumping and other forms of movement represent different activity contexts, but the present framework does not treat these modalities as the primary scientific objects. Instead, it considers how movement is organized and how that organization may influence the internal response generated by the organism.

A further distinction concerns functional outcome. Successful completion of a motor task demonstrates that a functional requirement has been met, but does not by itself establish how the organism achieved that outcome or whether the underlying organization remains stable, adaptive or physiologically sustainable over time. Following neurological injury, motor improvement may reflect either recovery of previously available motor functions or compensation through alternative movement strategies; these processes are conceptually distinct and may occur at neuronal, motor-performance and functional levels (Levin et al., 2009). Accordingly, the present framework distinguishes functional performance from movement organization, and both from the subsequent physiological response and adaptation.

The same principle applies to physiological and clinical measurements. Structural imaging, laboratory analyses, cardiovascular testing, electromyography and movement analysis each provide valuable information about specific dimensions of the organism. Their results should not be interpreted as interchangeable or as complete representations of whole-organism adaptive state. The present framework therefore does not seek to replace established measurements, but to investigate whether their temporal relationships, together with movement characteristics, can provide additional information about dynamic adaptation.

The temporal dimension is central to this question. An organism may respond acutely to a movement stimulus and subsequently return toward its prior state, or repeated exposure may progressively alter subsequent responses. Acute response, recovery and longer-term adaptation therefore represent distinct analytical levels. A single measurement before and after movement cannot by itself capture this trajectory. The proposed framework consequently considers a sequence of current state, movement stimulus, multisystem response, recovery, repeated exposure and adaptation (Iellamo, 2024; Furrer & Handschin, 2024).

Based on this background, the present paper proposes an integrated framework in which the organism is treated as a continuously adapting biological system and movement as both an observable expression of its current organization and a recurrent perturbation capable of modifying that state. The framework follows the sequence of current biological state, movement impulse, organization of movement, multisystem response, recovery or remodelling, repeated exposure and adaptation, followed by changes in functional and health trajectory.

The principal research question is:

How does the organization of movement influence the dynamic multisystem response and adaptive trajectory of the whole organism, and can measurable characteristics of this relationship contribute to the prevention of noncommunicable diseases?

The remainder of the paper is organized as follows. Section 2 presents the conceptual framework and positions it relative to existing approaches to whole-organism integration and movement science. Section 3 describes the methodological and measurement framework proposed for empirical evaluation. Section 4 defines the research propositions and falsifiability criteria. Section 5 discusses the potential scientific and preventive implications, limitations and priorities for future empirical research.

2. Existing Scientific Landscape and Conceptual Framework

2.1. Whole-organism physiology as an established scientific perspective

Contemporary biology does not treat the human organism exclusively as a collection of independent organs or isolated physiological systems. Systems biology explicitly emphasizes the study of the structure and dynamics of cellular and organismal function and the emergent properties of biological systems (Kitano, 2002).

Integrative physiology extends this perspective by examining how processes operating across multiple physiological levels contribute to whole-body function. A recent conceptual framework by Kuang (2026) further emphasizes that many physiological functions arise from coordinated causal processes extending across multiple levels of organization and proposes a five-domain framework intended to provide a coherent and operational representation of whole-body function.

Accordingly, the present framework does not propose whole-organism integration as a new scientific concept. Rather, it accepts organismal integration as established scientific territory and addresses a more specific problem concerning the relationship between movement and the resulting dynamic organization of the organism.

2.2. Dynamic physiological interaction and Network Physiology

Network Physiology provides one of the most direct existing approaches to the dynamic interaction of physiological systems. Bartsch et al. (2015) developed a system-wide approach for quantifying dynamic interactions among physiological systems using long-term, continuous, multichannel recordings and demonstrated that patterns of network organization are associated with physiological states and functions.

Network Physiology of Exercise subsequently extended this perspective specifically to exercise-related phenomena. Balagué et al. (2020) proposed a research programme focused on interactions and principles of coordination and integration among diverse physiological systems across spatial and temporal scales, including analysis of dynamic coupling, synchronization and multivariate physiological time series. Balagué et al. (2022) further developed methodological implications for dynamic couplings, physiological interactions and network-based exercise research.

These approaches are directly relevant to Adaptive Movement Research, but they address a distinct primary question. Network Physiology generally begins from physiological signals and seeks to characterize interactions among systems. The present framework instead asks whether the organization of the movement stimulus itself contributes to the formation of the subsequent multisystem response. The proposed relationship can therefore be represented as movement organization → biological perturbation → multisystem response → adaptive trajectory rather than treating movement only as a contextual condition under which physiological networks are observed.

2.3. Movement research and the concept of movement quality

Movement quality is already an established area of investigation, but its operationalization remains heterogeneous. Pfluegler et al. (2025) identified ten movement-quality domains in a scoping review of motor control and movement quality: flexibility, stability, accuracy, symmetry, speed, smoothness, adaptability, variability, efficiency and coordination. The review organized these domains within an ICF-based conceptual framework rather than establishing a universal movement-quality standard.

Wijekulasuriya et al. (2025) systematically reviewed movement-quality assessments in athletic populations and identified substantial variation in developmental approaches, included movements, scoring systems and reliability assessment. Thirty-six assessment approaches comprising 59 movements were identified, and only 59% of included studies reported reliability statistics. Wijekulasuriya et al. (2026) subsequently used a three-stage Delphi study to establish expert consensus on components of an athletic movement-quality assessment, while explicitly identifying reliability and validity as priorities for future work.

These studies have two implications for the present framework. First, movement quality should not be treated as a single universally accepted variable. It is better understood as a multidimensional description of observable movement organization whose interpretation depends on task, context and purpose of measurement. Second, the present research does not attempt to create another movement-quality scoring system. The central question is whether characteristics of movement organization have consequences for the biological stimulus experienced by the whole organism and for the subsequent multisystem response.

Thus, within the present framework, movement quality is an observable characteristic; it is not the final explanatory construct.

2.4. Physical activity and noncommunicable disease prevention

Physical activity has an established role in the prevention and management of NCDs. The World Health Organization defines physical activity broadly as bodily movement produced by skeletal muscles that requires energy expenditure and identifies regular physical activity as an important contributor to the prevention and management of cardiovascular disease, cancer, diabetes and other adverse health outcomes (World Health Organization, 2024).

This evidence provides the public-health foundation for Adaptive Movement Research. The present framework does not challenge the established value of physical activity and does not seek to replace conventional dose-related concepts such as frequency, duration or intensity.

Instead, it introduces a different question: Does the biological effect of movement depend only on how much movement occurs, or also on how the movement is organized by the individual organism?

2.5. From movement modality to movement stimulus

The conventional classification of movement commonly emphasizes the activity or modality being performed: walking, running, swimming, resistance activity, sport-specific activity, rehabilitation activity and so forth. The present framework proposes a different analytical level.

A movement modality describes what is being done. A movement impulse refers to the temporally bounded experimental episode, whereas a movement stimulus refers to the measurable biological perturbation associated with that episode. The same nominal activity can therefore produce different internal demands depending on how the movement is organized, the individual's current biological state and the history of previous exposure.

The framework consequently separates movement modality from external stimulus, movement organization and internal multisystem response. This distinction permits investigation of movement independently of the disciplinary context in which it is conventionally classified.

2.6. Movement as both output and perturbation

A central proposition of the present framework is that movement has a dual status. First, movement is an observable output of an already active biological system. Its execution reflects the current interaction of sensory information, neural processing, motor activation, musculoskeletal mechanics and physiological conditions.

Second, movement is a perturbation of that system. For the purposes of this framework, the internal biological environment refers to the dynamic state of interacting physiological processes, including mechanical, neural, cardiorespiratory, metabolic, autonomic and other relevant domains. Once initiated, movement changes this internal environment through altered mechanical loading, metabolic demand, cardiovascular and respiratory requirements, sensory input and other physiological conditions.

The relationship can therefore be represented as:

internal state → movement → altered internal state → subsequent response

The significance of this distinction is that the same movement can be viewed from two complementary directions: What does the movement reveal about the current organization of the organism? And what does the movement do to the subsequent organization of the organism? Adaptive Movement Research is concerned with the relationship between these two directions.

2.7. Functional outcome versus organization of function

Successful completion of a movement task is not equivalent to demonstrating a particular organization of the underlying biological processes. Two individuals may achieve the same external task outcome while using different movement strategies, different distributions of mechanical load or different neuromuscular organization. Similarly, after injury or other alterations in biological state, functional performance may improve through acquisition of a new or compensatory movement strategy rather than through exact restoration of a previous movement pattern (Levin et al., 2009).

The framework does not interpret compensation as inherently pathological. A compensatory strategy may be functionally adaptive under particular conditions. The relevant research question is whether repeated use of a particular strategy produces a reproducible physiological trajectory and whether that trajectory is favourable, neutral or potentially adverse relative to the prespecified outcome and study context.

2.8. Static measurements and dynamic biological trajectories

Conventional clinical and physiological measurements provide highly valuable information, but each measurement captures a particular domain. Structural imaging provides information about anatomy and tissue structure. Laboratory measurements characterize selected biochemical or physiological variables. Cardiovascular and respiratory tests characterize specific functional responses. Electromyography provides information about electrical muscle activity. Movement analysis provides observable information about motor behavior.

The present framework does not imply that these measurements are insufficient for their intended purposes. Instead, it distinguishes measurement of a domain from representation of the organism's dynamic adaptive trajectory.

At the same time, this interpretation should not be read as evidence that whole-organism integration is absent from contemporary science. Systems biology, integrative physiology and Network Physiology already provide established frameworks for studying integrated biological function and dynamic inter-system interactions (Kitano, 2002; Bartsch et al., 2015; Balagué et al., 2020; Kuang, 2026).

Accordingly, Adaptive Movement Research proposes that the temporal relationship among multiple measurable variables may be more informative for some research questions than any single measurement considered independently.

2.9. The conceptual gap addressed by the present framework

The existing literature therefore provides substantial knowledge at each major level: physical activity and NCD prevention are well established; multisystem physiological responses to movement are established; Network Physiology provides methods and theory for dynamic physiological integration; movement-quality research provides multiple domains for

characterizing movement organization; and integrative physiology continues to develop frameworks for whole-body functional organization.

The potential gap addressed by Adaptive Movement Research is narrower. The question is whether these bodies of knowledge can be connected through a common experimental framework in which movement organization → systemic perturbation → multisystem response → recovery → repeated exposure → adaptation is treated as one temporally structured research process.

This framework does not assume that such a relationship is necessarily strong, causal or clinically useful. Its purpose is to make the relationship explicit enough to be empirically tested.

2.10. Position of Adaptive Movement Research

Adaptive Movement Research therefore occupies an interdisciplinary position rather than constituting a new medical specialty or movement modality. The programme integrates perspectives from physiology, systems biology, Network Physiology, movement science and preventive medicine.

Its fundamental object is neither exercise, sport, fitness, rehabilitation nor a particular movement technique. The central object is the relationship between movement and the dynamic organization of the whole organism.

The long-term applied question is whether understanding this relationship can provide additional information relevant to the preservation of health and the prevention of NCDs.

The framework consequently positions movement at the intersection of observable behaviour → biological perturbation → multisystem physiological response → adaptation → health trajectory.

3. Methodological and Measurement Framework

3.1. Methodological position

The present paper develops a conceptual and methodological framework rather than reporting an original experimental study. Its purpose is to define how the proposed relationship between movement organization, multisystem response and adaptation could be investigated empirically.

3.2. Definition of the movement impulse

For the purposes of this framework, the term movement impulse is introduced as an operational designation for a temporally bounded episode of movement used to produce and characterize a measurable perturbation of the organism.

The impulse is characterized independently of its conventional classification. Its relevant properties may include duration, amplitude, velocity, acceleration, mechanical demand, temporal structure, repetition, direction, task constraints and the individual's initial biological state.

The framework therefore distinguishes movement modality, external stimulus, movement organization and internal response. This separation prevents the experimental question from becoming dependent on a particular movement category.

3.3. Baseline biological state

Before a movement stimulus is applied, the individual's current state should be characterized to the extent required by the research question. Potential baseline domains include demographic and anthropometric characteristics, relevant medical and functional status, baseline movement characteristics, cardiorespiratory measures, autonomic measures, selected metabolic or biochemical variables, relevant symptoms, recent activity or loading history and recovery status where measurable.

The purpose is not to create a universal diagnostic panel. Baseline variables should be selected according to the specific research question and the hypothesized mechanisms under investigation.

3.4. Measurement of movement organization

Movement organization should be treated as multidimensional rather than represented by a single score. Candidate measurement domains include spatial organization, temporal organization, mechanical organization and neuromuscular organization.

Spatial measures may include joint and segment trajectories, range of motion, symmetry and inter-segment relationships. Temporal measures may include event onset and offset, movement duration, phase relationships, sequencing and temporal variability. Mechanical measures may include external forces, loading distribution, acceleration, joint kinetics where available and relationships between movement and external mechanical interaction. Neuromuscular measures may include muscle activation timing, amplitude, co-activation and inter-muscular relationships where appropriate.

These domains are consistent with existing movement-quality research, but their purpose in the present framework differs from conventional movement scoring. The objective is not to assign a normative quality score but to characterize the organization of the movement stimulus that precedes the systemic response (Swain et al., 2023; Pfluegler et al., 2025; Wijekulasuriya et al., 2025).

3.5. Temporal organization as a primary measurement dimension

The temporal relationship among movement and physiological events is a central methodological element of the framework.

- Movement-event timing — onset, offset and duration.
- Inter-segment sequencing — temporal order of movement contributions by different segments.
- Relative phase relationships — temporal relationships between oscillatory or cyclic components.
- Neuromuscular timing — temporal relationship between muscle activation and movement.
- Movement-response latency — delay between the movement stimulus and a measurable physiological response.
- Cross-system lag structure — temporal relationships between physiological signals.
- Recovery kinetics — rate and pattern of return toward baseline or transition to a new state.

The purpose of these measures is not to assume that one particular temporal pattern is universally optimal. Rather, temporal characteristics provide a means of testing whether differences in movement organization are associated with differences in subsequent multisystem response.

3.6. Mechanical and neuromuscular response

Where experimentally justified, movement analysis may be combined with force measurement and neuromuscular recording. Surface electromyography can provide information about electrical activity in selected muscles, while force plates, dynamometry or other mechanical systems can characterize external forces and mechanical interactions.

The framework explicitly distinguishes electrical activation from mechanical contribution. An EMG signal alone does not establish the amount of force produced, the mechanical work performed, the distribution of load across other tissues or the contribution of a muscle to the whole-organism response.

3.7. Multisystem physiological response

The physiological response should be assessed at multiple levels where justified by the research question. Candidate domains include cardiorespiratory response, autonomic response, metabolic response, endocrine response, immune/inflammatory response, symptom response and system-level interactions. The appropriate measurement set should follow the research question, hypothesized mechanism, feasibility and measurement reliability. For the purposes of an individual study, a multisystem response is defined as a temporally characterized response profile involving prespecified variables from more than one physiological domain relevant to the research question. This multidimensional approach is consistent with systematic evidence that acute responses to movement/exercise may involve multiple interacting dimensions rather than a single outcome variable (Morava et al., 2024).

This approach is consistent with Network Physiology, which uses multichannel physiological recordings to characterize dynamic interactions among physiological systems and their relation to physiological states (Bartsch et al., 2015; Balagué et al., 2020; Balagué et al., 2022). The framework does not require simultaneous measurement of all listed domains; empirical studies should use hypothesis-driven measurement subsets with prespecified primary response domains.

3.8. Dynamic multimodal synchronization

A central methodological requirement is temporal synchronization of measurements. Movement data, neuromuscular activity, mechanical signals and physiological signals should be aligned to a common temporal reference whenever the research question concerns sequence, latency, coupling or recovery.

A conceptual data structure is therefore: Movement(t), Neuromuscular(t), Mechanical(t), Cardiorespiratory(t), Autonomic(t), Metabolic/Biochemical(t) and Symptom(t), rather than separate measurements collected independently and compared only at baseline and endpoint.

Wearable and multisensor technologies make increasingly continuous measurement possible, while systematic review evidence indicates that multisensor configurations can provide a broader range of movement-related variables than single-sensor approaches (Swain et al., 2023).

3.9. Acute response

The first analytical interval is the acute response to an individual movement stimulus. The objective is to characterize the magnitude, timing and distribution of changes following the stimulus rather than to interpret an immediate response as adaptation.

- Onset latency.
- Peak response.
- Response duration.
- Change from baseline.
- Cross-system temporal relationships.
- Symptom response.
- Recovery onset.

An acute response constitutes an observation of how the organism reacts to a defined movement perturbation. It does not establish whether that response is beneficial, harmful or adaptive in the longer term.

3.10. Recovery and resolution

Recovery is treated as an analytically distinct phase. The relevant question is not simply whether a measured variable returns to its baseline value. Recovery may involve different rates and patterns across physiological systems, and a variable may transition toward a new steady state rather than return exactly to the original level.

Candidate recovery measures include time to predefined recovery criteria, recovery slope, delayed or secondary responses, synchronization or desynchronization of system-level responses and persistence of movement or symptom changes.

3.11. Repeated exposure and adaptive trajectory

A repeated-exposure study should not rely exclusively on a baseline-to-final comparison. The preferred analytical structure is: Stimulus 1 → Response 1 → Recovery 1; Stimulus 2 → Response 2 → Recovery 2; ...; Stimulus n → Response n → Recovery n.

Adaptation would therefore be operationalized not simply as an improvement in a selected outcome, but as a reproducible change in response characteristics across repeated exposures. Potential indicators include altered response magnitude, altered response latency, altered recovery kinetics, altered movement organization, altered cross-system relationships, increased tolerance to the stimulus and persistent changes in functional characteristics.

The precise criteria for a favourable trajectory relative to a prespecified outcome must remain hypothesis-specific rather than assumed in advance (Iellamo, 2024; Furrer & Handschin, 2024).

3.12. Individual state dependence

The framework assumes that response cannot be interpreted independently of the individual's current state. The same movement stimulus may produce different responses in individuals who differ in baseline physiology, previous exposure, functional status, symptoms, disease state, sleep, recovery or other relevant factors.

Accordingly, individualization should not be based solely on demographic categorization or predefined labels. Where possible, it should be informed by repeated measurement of the individual's own movement and physiological response.

3.13. Response-guided modulation

A future experimental implementation may use observed response characteristics to modify the next movement stimulus. The conceptual sequence is baseline state → movement stimulus → response profile → reassessment → modified stimulus → subsequent response.

Potential decision variables may include movement organization, mechanical demand, temporal characteristics, amplitude, duration, repetition or recovery interval. The purpose is not to create a universal movement prescription algorithm. Instead, the research question is whether response-guided modification produces a distinguishable and reproducible adaptive trajectory compared with a fixed-stimulus approach.

3.14. Multimodal integration and analytical strategy

The framework requires analysis at two complementary levels. The first level characterizes each measurement domain independently. The second level examines temporal relationships among domains.

Candidate analytical approaches include time-series analysis, phase and coupling measures, change-point analysis, repeated-measures modelling, multilevel models and other methods appropriate to the structure of the collected data. The specific statistical or dynamical method should be selected after the measurement design is defined.

3.15. Measurement limitations

- No single sensor or physiological measurement directly represents the state of the whole organism.
- Different measurements operate at different spatial and temporal scales.
- Measurement error may create apparent changes unrelated to biological adaptation.
- Observed correlations between movement and physiological variables do not establish causal direction.
- Changes after repeated exposure may reflect learning, fatigue, compensation, behavioural change or other processes rather than beneficial adaptation.
- Wearable and non-laboratory measurements may have lower precision than laboratory-grade instrumentation, although they may provide greater ecological validity and repeated sampling.

Accordingly, validation should proceed from controlled laboratory measurements toward increasingly naturalistic observation rather than assuming equivalence between all measurement environments.

3.16. Proposed empirical progression

The framework suggests a staged empirical programme.

- Stage 1 — Measurement feasibility: establish whether movement organization and relevant physiological variables can be measured synchronously and reproducibly.

- Stage 2 — Acute response: determine whether systematically different movement organizations at comparable external stimulus produce distinguishable multisystem response profiles.
- Stage 3 — Recovery: characterize recovery kinetics and temporal relationships among response domains.
- Stage 4 — Repeated exposure: determine whether response characteristics change systematically across repeated exposure.
- Stage 5 — Adaptive trajectory: determine whether persistent changes can be distinguished from acute variation, learning effects and measurement noise.
- Stage 6 — Health relevance: examine associations with established health-related outcomes such as blood pressure, glycemic control, inflammatory or metabolic markers, cardiorespiratory fitness or functional capacity.
- Stage 7 — Longitudinal prevention: examine whether validated movement-response trajectories are associated with incident NCD outcomes or changes in established NCD risk profiles in appropriately selected populations.

The staged design deliberately separates mechanistic validation from clinical and preventive inference. Evidence of a relationship between movement organization and acute multisystem response would not by itself justify claims concerning long-term health or disease prevention.

3.17. Methodological endpoint

The methodological objective of the framework is therefore not to identify a universally correct movement pattern. It is to establish whether the following relationship can be measured and tested: movement organization → multisystem response → recovery → repeated response → adaptation → health trajectory.

The principal empirical test is whether variation in movement organization contributes measurable information about systemic response and adaptive trajectory that is not explained adequately by external activity dose, conventional functional performance or isolated physiological measurements. If this relationship cannot be demonstrated reproducibly, the proposed framework must be revised or rejected.

4. Research Propositions and Falsifiability Criteria

4.1. Purpose of the propositions

The integrated framework developed in the preceding sections is intended to generate empirically testable propositions rather than establish a new biological theory. Each proposition therefore defines a relationship that can, in principle, produce supporting, inconclusive or contradictory observations.

The propositions are organized sequentially from the immediate relationship between movement organization and systemic response to longer-term adaptation and, ultimately, preventive relevance.

4.2. Proposition 1 — Movement organization and multisystem response

P1: At a comparable external movement stimulus, systematic differences in the temporal and organizational characteristics of movement will be associated with reproducible differences in multisystem physiological response.

For empirical testing, movement organization may be characterized by movement-event timing, inter-segment sequencing, relative phase relationships, temporal variability and neuromuscular timing. Systemic response may be represented by temporally aligned cardiovascular, respiratory, autonomic, metabolic or other hypothesis-specific signals.

Support for P1 would require reproducible differences in prespecified response characteristics such as response magnitude, onset latency, temporal coupling or recovery initiation. A finding that movement organization has no reproducible association with systemic response under appropriately controlled conditions would weaken P1.

4.3. Proposition 2 — Systemic response and adaptive trajectory

P2: Repeated exposure to a defined movement stimulus will be associated with systematic within-person changes in prespecified characteristics of the multisystem response and recovery trajectory.

Relevant longitudinal characteristics may include response magnitude, response latency, recovery kinetics, variability of repeated responses, temporal coupling between physiological systems, persistence of changes after recovery and changes in movement organization itself.

Support for P2 would require evidence that the response trajectory changes systematically beyond expected measurement error, acute day-to-day variability or simple learning effects.

4.4. Proposition 3 — Incremental information from temporal organization

P3: Temporal characteristics of movement organization and their temporal relationships with physiological signals will provide incremental information about adaptive state beyond conventional measures of activity dose, functional performance and isolated physiological measurements.

Candidate variables include movement-event timing, inter-segment sequencing, relative phase relationships, temporal variability, movement-to-response latency, cross-system coupling and lag structure, and recovery kinetics.

Incremental information should be assessed using prespecified nested-model comparisons rather than statistical significance alone. For continuous outcomes, incremental explained variance may be reported, but the minimum magnitude considered meaningful should be prespecified according to the outcome, study design, expected measurement error and scientific or clinical relevance. For prediction models, incremental value should additionally be evaluated using appropriate measures of discrimination, calibration, likelihood-based comparison and out-of-sample performance. AIC/AICc may be reported where applicable. Reclassification measures may be considered as secondary analyses when established risk categories are clinically prespecified. These requirements are intended as study-design principles rather than universal biological thresholds (Collins et al., 2024).

P3 will be considered supported only when movement-derived temporal information demonstrates a reproducible and practically meaningful improvement in explanation or prediction across prespecified analyses and validation procedures.

4.5. Proposition 4 — Response-guided modulation (downstream experimental hypothesis)

P4: When reproducible movement-response patterns can be identified, modification of a subsequent movement stimulus according to prespecified response criteria will produce a distinguishable subsequent response trajectory compared with a fixed-stimulus approach.

A future controlled design could compare a fixed-stimulus condition with a response-guided condition in which the subsequent movement stimulus is modified according to prespecified characteristics of movement organization, physiological response, recovery or symptom response.

Support for P4 would require that the response-guided condition produce a reproducible difference in a prespecified trajectory measure rather than simply a transient difference in one acute variable.

4.6. Proposition 5 — Health relevance as a downstream hypothesis

P5: Validated movement-response trajectories will show reproducible associations with established physiological, functional or behavioural risk factors relevant to NCD development.

An illustrative future study could recruit an appropriately selected cohort of adults with elevated but not necessarily established cardiometabolic risk. Candidate outcomes may include resting and ambulatory blood pressure, glycemic control including HbA1c where appropriate, cardiorespiratory fitness or functional capacity, body-composition-related measures, and selected inflammatory or metabolic biomarkers.

The choice of outcomes should be defined before analysis and correspond to the disease-risk mechanism being studied. P5 would require reproducible associations after appropriate adjustment for conventional risk factors, baseline health state and relevant behavioural exposures.

4.7. Proposition 6 — Longitudinal preventive hypothesis

P6: If validated movement-response trajectories are associated with established NCD risk factors, prospective longitudinal studies will determine whether these trajectories are associated with subsequent incident NCD outcomes or changes in composite disease-risk profiles.

An illustrative longitudinal population could consist of adults with elevated cardiometabolic risk but without the incident outcome under investigation at baseline. Potential primary outcomes could include prespecified incident cardiovascular disease, type 2 diabetes or clinically relevant composite cardiovascular outcomes. Secondary outcomes could include longitudinal changes in blood pressure, HbA1c, lipid profile, cardiorespiratory fitness or functional capacity.

P6 represents a future longitudinal research objective rather than a claim of the present paper. The framework does not assume that movement organization independently causes prevention of NCDs.

4.8. Hierarchical interpretation of the propositions

The propositions form a cumulative research sequence:

$$\mathbf{P1 \rightarrow P2 \rightarrow P3 \rightarrow P4 \rightarrow P5 \rightarrow P6}$$

Failure at an earlier stage would require reconsideration of downstream propositions. Evidence for P1 and P2 would not by itself justify P5 or P6.

4.9. Falsifiability criteria

- Criterion 1 — No movement-organization effect: if controlled studies demonstrate no reproducible differences in multisystem response under comparable external demand, P1 is not supported.
- Criterion 2 — Acute response without adaptation: if movement organization influences acute response but repeated exposure does not produce distinguishable trajectories, the adaptive extension is not supported.
- Criterion 3 — No incremental value: if movement-organization variables provide no practically meaningful information beyond activity dose, functional measures and established physiological variables, the additional analytical layer is weakened. The minimum acceptable improvement should be prespecified for the specific outcome and design rather than adopted as a universal threshold.
- Criterion 4 — No response-guided effect: if response-guided modification produces no reproducible change in trajectory compared with a matched fixed-stimulus approach, the feedback component requires revision.
- Criterion 5 — No health relevance: if validated movement-response trajectories show no reproducible association with prespecified health-related outcomes or NCD risk factors, the preventive application is not supported.
- Criterion 6 — Existing frameworks provide complete explanation: if established approaches adequately explain the proposed relationship without the additional framework developed here, the conceptual contribution of Adaptive Movement Research must be reconsidered.

4.10. Causal interpretation

The propositions deliberately distinguish association from causation. A temporal relationship between movement organization and physiological response may provide evidence consistent with a mechanistic relationship, but temporal precedence alone does not establish causal direction.

- Baseline physiological differences.
- Fatigue.
- Learning effects.
- Pain or symptom variation.
- Psychological state.
- Environmental conditions.
- Measurement error.
- Unmeasured changes in external workload.

- Biological confounding.
- Regression to the mean.

Accordingly, causal interpretation will require progressively stronger experimental and longitudinal designs.

4.11. Interpretation of negative findings

Negative findings should not automatically be considered failure of the entire research programme. Evidence that a particular movement characteristic does not predict systemic response may refine the framework by excluding that characteristic. Likewise, evidence that movement organization contributes only under particular physiological states may indicate that the relationship is state-dependent rather than universally applicable.

4.12. Relationship to the overall objective

The ultimate objective of Adaptive Movement Research is not to classify movement as good or bad. It is to determine whether the organization of movement can be understood as a measurable component of the biological stimulus experienced by the whole organism, whether that stimulus produces identifiable multisystem responses and adaptive trajectories, and whether these trajectories may eventually have relevance to health preservation and NCD prevention.

5. Discussion, Implications, Limitations and Future Research

5.1. Principal contribution of the framework

The present paper proposes an integrated research framework for investigating movement as both an observable expression of ongoing whole-organism organization and a recurrent perturbation capable of modifying the internal state of the organism.

The framework does not propose that whole-organism integration is absent from contemporary science. Systems biology, integrative physiology and Network Physiology already provide established approaches for examining organismal organization, multisystem interactions and dynamic physiological states (Kitano, 2002; Bartsch et al., 2015; Balagué et al., 2020; Balagué et al., 2022). A recent framework by Kuang (2026) further proposes an operational representation of whole-body function across multiple functional domains.

The proposed contribution is therefore narrower: the formulation of a research structure in which the organization of movement is examined as a potential determinant of the biological stimulus experienced by the organism and is then related temporally to multisystem response, recovery, repeated exposure and adaptation.

The framework does not introduce a new physiological mechanism. Rather, it proposes a way of connecting existing areas of knowledge through a common experimental sequence: movement organization → multisystem response → recovery → repeated exposure → adaptation → health trajectory.

5.2. Relationship to existing scientific frameworks

The framework should be understood as complementary to, rather than competitive with, existing fields. Systems biology provides the broader basis for studying biological systems as interacting dynamic entities. Integrative physiology addresses processes extending across

anatomical and physiological levels. Network Physiology provides quantitative approaches for measuring dynamic interactions among physiological systems, including during exercise (Kitano, 2002; Bartsch et al., 2015; Balagué et al., 2020; Balagué et al., 2022).

The present framework begins from a different analytical point. Instead of beginning with physiological signals and asking how they interact, it begins with a movement stimulus and asks how the organization of that stimulus relates to the subsequent organization of the systemic response.

Existing network-oriented perspective: physiological signals → inter-system interactions → system-level state.

Present research perspective: movement organization → biological perturbation → multisystem response → adaptation.

The two approaches are not mutually exclusive. Network Physiology may provide important analytical methods for evaluating the systemic response proposed in the present framework.

5.3. Movement as a biological perturbation

A central implication of the framework is that movement should not be considered exclusively as an outcome of motor control or as a category of physical activity. Movement is produced by an already active biological system. Once initiated, however, the movement changes the conditions under which those same systems operate.

This creates a recursive relationship: internal state → movement → altered internal state → subsequent response.

Movement has particular research value because it is recurrent, observable, quantifiable and naturally embedded in everyday human behaviour. The established health effects of physical activity provide the public-health background for this proposition (World Health Organization, 2024).

5.4. Movement quality and movement organization

Movement-quality research already encompasses multiple domains, including coordination, stability, symmetry, variability, adaptability, smoothness and efficiency. The diversity of these domains demonstrates that movement quality cannot reasonably be reduced to a single universal score (Pflueger et al., 2025; Wijekulasuriya et al., 2025; Wijekulasuriya et al., 2026).

The present framework therefore treats movement quality as an observable multidimensional characteristic rather than as a predetermined normative standard.

The significance of movement organization should ultimately be determined by its relationship to the biological response and adaptive trajectory of the individual. This represents an empirical question rather than a normative rule.

5.5. Functional success versus biological adaptation

A person may complete a functional task through different movement strategies. The external result may therefore remain stable while the organization of the underlying movement changes. Following neurological injury or other changes in biological state, functional improvement may

also involve compensation through an alternative motor strategy rather than exact restoration of the previous movement organization (Levin et al., 2009).

The framework does not interpret compensation as inherently pathological. A compensatory strategy may be functionally adaptive under particular conditions. The relevant research question is whether repeated use of a given movement organization is associated with a reproducible physiological trajectory and whether that trajectory is favourable, neutral or potentially adverse relative to the prespecified outcome.

5.6. Static measurements versus adaptive trajectories

MRI, laboratory testing, cardiovascular assessment, spirometry, pulse oximetry, electromyography and movement analysis each provide valuable information within their intended measurement domains. The framework does not imply that any of these methods is inadequate for its specific purpose.

The methodological problem is different: a single measurement cannot necessarily represent the entire temporal organization of a dynamic biological system.

Network Physiology and integrative physiology provide an established basis for examining dynamic interactions among physiological systems across multiple spatial and temporal scales (Bartsch et al., 2015; Balagué et al., 2020; Kuang, 2026). The potential contribution of movement therefore lies not in replacing structural or physiological measurements, but in adding a repeatedly observable functional signal that can be examined together with those measurements.

5.7. Implications for measurement and experimental design

Future studies should move beyond isolated pre- and post-intervention measurements where the research question concerns temporal organization. Where technically feasible, movement, neuromuscular, mechanical and physiological signals should be synchronized to allow evaluation of event timing, latency, relative phase, coupling and recovery.

The appropriate level of measurement will depend on the specific hypothesis. A study investigating movement organization and cardiovascular response may not require endocrine or immune measurements, whereas a study investigating broader systemic adaptation may require broader multisystem sampling.

Repeated measures of the same individual may be particularly informative. Between-person comparisons can reveal population-level associations, but within-person trajectories may provide a more direct view of how the organism responds to repeated perturbation.

5.8. Potential relevance to preventive medicine

The long-term objective of Adaptive Movement Research is prevention of NCDs, but the present framework does not claim that movement organization has already been validated as a preventive biomarker.

Physical activity already has an established role in prevention of cardiovascular disease, diabetes and several cancers. WHO also identifies raised blood pressure, elevated blood glucose, abnormal blood lipids and overweight/obesity as major metabolic risk factors associated with NCDs (World Health Organization, 2024; World Health Organization, 2025).

Any future movement-derived assessment would be intended to complement, not replace, established clinical risk assessment and diagnostic procedures. Its potential value would depend on demonstrating reproducible incremental information beyond conventional measures.

A possible future pathway is: movement organization → multisystem response → adaptive trajectory → established health-related markers → NCD risk. Only if this pathway is supported empirically would it be appropriate to investigate whether movement-response characteristics have value in preventive interventions or longitudinal risk assessment.

5.9. Potential application across research domains

Although the framework deliberately excludes exercise, sport, fitness, wellness and rehabilitation as fundamental scientific categories, it may eventually be tested within each of these contexts. Potential study populations include healthy adults, older adults, individuals with elevated cardiometabolic risk, individuals undergoing rehabilitation, preoperative populations, athletes and populations with established chronic disease.

The scientific object would remain unchanged: movement organization → systemic response → adaptation. Only the population, experimental context and health outcomes would differ.

5.10. Limitations

- The present paper is conceptual and contains no original experimental data.
- The framework may overlap substantially with existing approaches once operationalized.
- Movement organization is multidimensional and context-dependent.
- Temporal association does not establish causality.
- Measurement error may mimic biological change.
- The framework does not yet specify a universal definition of favourable adaptation.
- The long-term NCD hypothesis is subject to substantial confounding.

These limitations define the methodological requirements for testing the framework.

5.11. Future empirical research

Future research should proceed hierarchically: reliable measurement; acute-response studies; recovery and repeated-exposure studies; adaptive-trajectory analysis; association with established health-related outcomes; and finally longitudinal investigation of incident NCD outcomes. Intervention studies could then determine whether modifying movement organization or subsequent movement stimuli according to observed response produces a favourable adaptive trajectory compared with predefined or fixed approaches.

This sequence is intentionally conservative. It prevents movement directly from a conceptual observation to a therapeutic or preventive claim without intermediate evidence.

5.12. Interdisciplinary research implications

The framework is inherently interdisciplinary because no single discipline contains all of the required measurement domains. Physiology provides the study of systemic regulation and adaptation. Systems biology provides approaches for understanding interactions among biological components. Network Physiology provides quantitative methods for dynamic multisystem relationships. Movement science provides methods for characterizing movement

organization. Preventive medicine provides the health and NCD framework within which long-term relevance can be evaluated.

5.13. Overall interpretation

The present framework should not be interpreted as a claim that movement is a universal diagnostic instrument, that every movement pattern has a unique systemic effect, or that qualitative movement alone determines health outcomes.

Its central proposition is narrower: the organization of movement may represent a potentially measurable component of the biological stimulus experienced by the whole organism, and the temporal relationship between movement organization, multisystem response and adaptation may contain information relevant to understanding long-term functional and health trajectories.

5.14. Conclusion

Adaptive Movement Research proposes an integrated framework in which movement is considered both an observable expression of ongoing whole-organism organization and a recurrent perturbation capable of modifying the internal biological environment.

The framework connects: current biological state → movement impulse → movement organization → multisystem response → recovery/remodelling → repeated exposure → adaptation → functional and health trajectory.

Its central scientific question is whether the organization of movement contributes measurable information about systemic response and adaptive trajectory beyond movement quantity, functional performance and isolated physiological measurements.

The framework does not establish a new biological law, diagnostic method or preventive intervention. It defines a sequence of testable research questions through which the relationship between movement, whole-organism adaptation and long-term health can be investigated.

If subsequent empirical studies demonstrate reproducible movement-response relationships, distinguishable adaptive trajectories and incremental health-relevant information, the framework could provide a basis for response-guided movement research and, ultimately, for investigating whether quality of movement has a meaningful role in the prevention of noncommunicable diseases.

Table 1. Conceptual distinctions used in the framework

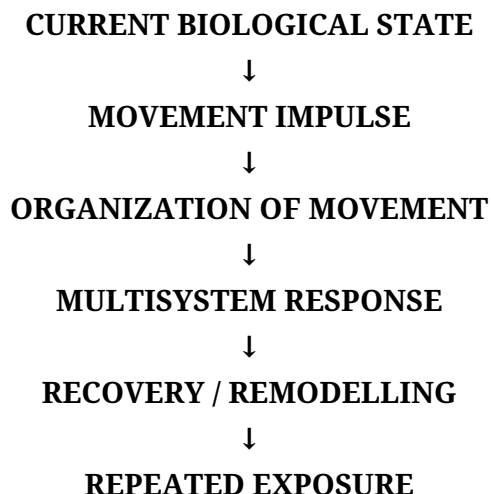
Concept	Primary question	Role in framework
Physical activity	How much and how often does a person move?	Exposure context
Movement quality	What observable characteristics describe movement?	Observable characteristics
Motor control	How is movement regulated?	Underlying mechanisms
Internal response	What changes physiologically?	Systemic response
Recovery	How does the system resolve the perturbation?	Transition
Adaptation	What changes with repeated	Long-term response

Concept	Primary question	Role in framework
	exposure?	
Functional state	What can the person do?	Observable outcome
Health trajectory	How does state change over time?	Long-term relevance
NCD prevention	How can adverse health trajectories be prevented?	Applied objective
Movement organization	How are movement characteristics temporally, spatially, mechanically and neuromuscularly organized?	Candidate explanatory level

Table 2. Evidence status and research hypotheses

Element	Status	Treatment in Paper 1
Physical activity and NCD prevention	Established	Scientific background
Multisystem response to movement	Established	Scientific background
Dynamic physiological integration	Established research domain	Scientific background / benchmark
Movement organization → systemic response	Hypothesis	P1
Systemic response → adaptive trajectory	Hypothesis	P2
Movement organization → incremental information	Hypothesis	P3
Response-guided movement modulation	Research hypothesis	P4
Trajectory → health-related outcomes	Downstream hypothesis	P5
Trajectory → NCD outcomes	Long-term hypothesis	P6

Figure 1. Integrated Research Framework





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